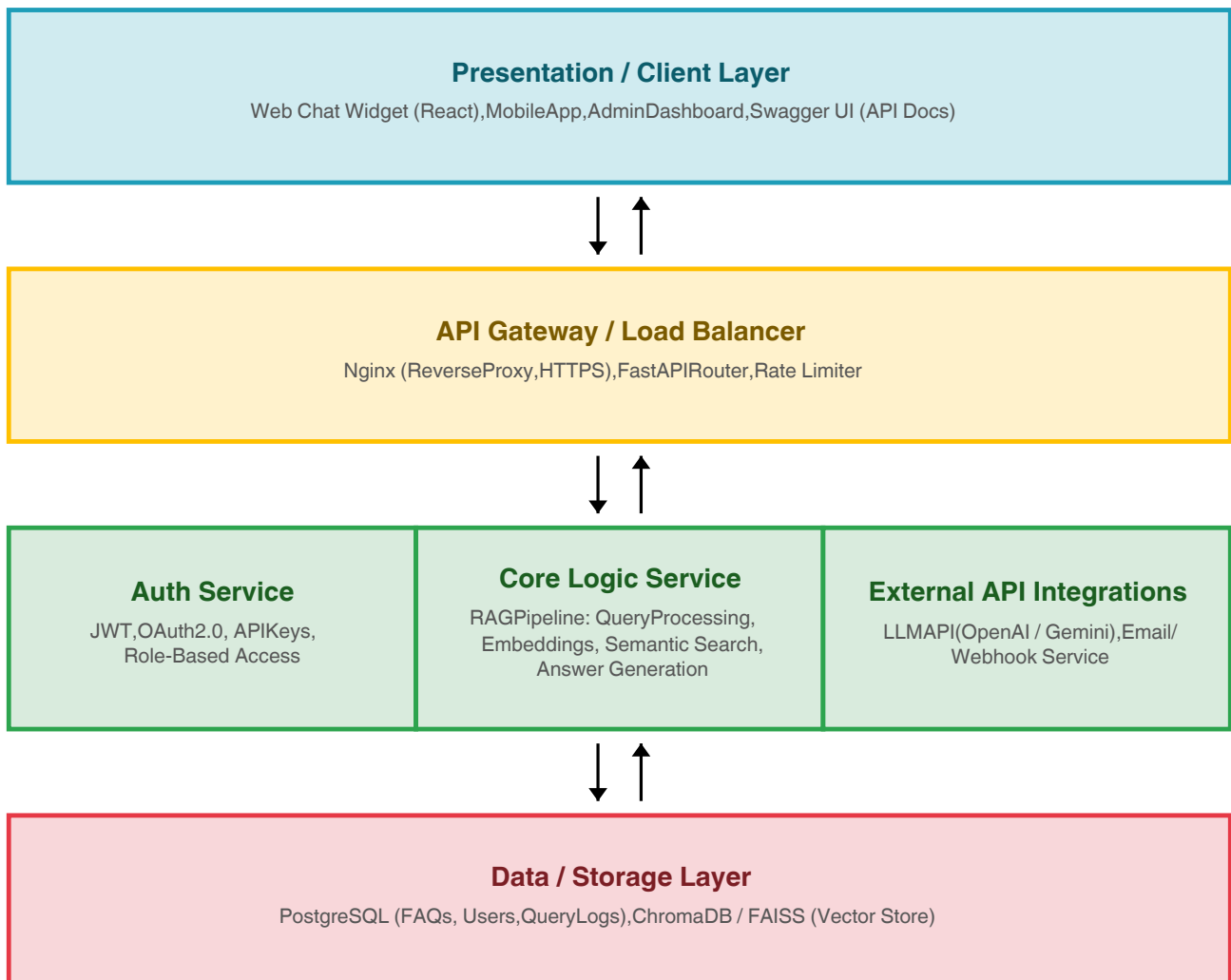


Solution Architecture

Date	08 October 2026
Team ID	UNM1611GUFRAN/SWTID-2026-9802
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Client-facing interfaces through which users and developers interact with the system: a chat widget / web app to ask questions, an admin dashboard to manage FAQs and view analytics, and Swagger UI for API documentation and testing.	React.js, HTML5, CSS3, Tailwind CSS, Swagger UI
API Gateway	Single entry point for all requests. Terminates HTTPS, routes calls to the right service, balances load, validates requests and enforces rate limits per API key.	Nginx, FastAPI, Uvicorn, Docker
Auth Service	Authenticates clients and admins and authorizes access by role (Admin, Editor, API Client). Issues, validates and revokes tokens and API keys.	JWT, OAuth 2.0, API Keys, Python
Core Logic Service (Microservice)	Runs the FAQ pipeline: cleans the user query, converts it to an embedding, retrieves the most similar approved FAQs, generates the answer with an LLM, attaches a confidence score and source, and applies a fallback for low-confidence queries.	Python, FastAPI, Sentence-Transformers, LangChain (optional)
External API Integrations	Connects to third-party services: the LLM / embedding API for answer generation and optional email or webhook services for notifications and escalations.	OpenAI / Google Gemini API, SMTP / Email Service, Webhooks
Database	Stores FAQ content, users, API keys, query logs and feedback in a relational database, and FAQ embeddings in a vector database for fast semantic search.	PostgreSQL, ChromaDB / FAISS, SQLAlchemy
Deployment & CI/CD	Containerized services deployed on cloud infrastructure with automated testing, build and deployment on every code push.	Docker, AWS (EC2 / ECS), GitHub, GitHub Actions